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RESEARCH ARTICLE

Enhancing Sound-Based Anomaly Detection Using Deep Denoising Autoencoder

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ABSTRACT Traditional sensor-based methods that detect machine malfunctions in industrial environments are often costly and complex; sound-based anomaly detection offers a simpler alternative. Such methods, however, must contend with industrial noise that masks the sound patterns necessary for effective diagnosis. This study proposes a method that uses a deep denoising autoencoder to filter out various levels of industrial noise from audio data and employs unsupervised learning models for rapid and accurate anomaly detection. The primary novelty of the proposed methodology lies in the audio data preprocessing techniques and the customized denoising process that is tailored to the noise levels of various industrial environments. Several experiments using different types of industrial machinery, such as pumps, valves, and slide rails, demonstrated the efficiency, effectiveness, and rapid processing capabilities of the proposed methodology. Specifically, the experimental results show that the proposed methodology afforded an average area under the curve performance improvement of approximately 18% compared to previous studies.

INDEX TERMS Acoustic signal processing, anomaly detection, automation, fault diagnosis, feature extraction, machine learning, machinery, noise, signal analysis, unsupervised learning.

I. INTRODUCTION

Advances in automation have increased the complexity of machinery in industrial settings, leading to an increased risk of machine malfunction. Such malfunctions can create safety hazards and quality concerns, reduce productivity, and disrupt production lines. These problems are highly significant in the industrial sphere; as a result, mechanical fault diagnosis technology has become a critical area of study [1]. Traditional methods typically rely on various sensors, such as vibration, temperature, and current sensors, to diagnose the condition of machinery [2]. Machines emit specific sound patterns when malfunctioning or when they develop anomalies or defects [3], [4], [5]. The collection of machinery-generated audio data that reveal patterns of concern not only reduces sensor costs but also eliminates the need to disassemble the machinery or the system. This approach aids the diagnosis of various malfunctions by reflecting the internal operational condition of the machinery, without interrupting production lines or facility operations. Consequently, many studies on mechanical fault diagnosis using sounds have been published recently, and several state-of-the-art approaches have been proposed [6], [7], [8].

However, some problems remain when sound is used to diagnose mechanical faults. Collection of abnormal machinery noises is often costly. Also, although most studies developed supervised learning models that label normal and abnormal data, the scarcity of abnormal data in real industrial settings complicates the practical implementation of such supervised learning models [9]. The complexity of real-world industrial environments, compounded by myriad ambient noises from various machines, further exacerbates these problems [10]. Such noises can obscure the sound patterns emitted by machines, adversely affecting failure detection performance [11], [12], [13]. Particularly in industrial settings, environmental sounds easily contaminate audio data, complicating the application of sound-based, mechanical fault diagnosis.

To overcome these problems, this study proposes an unsupervised learning-based anomaly detection methodology that effectively diagnoses mechanical faults in audio data containing various levels of industrial noise. A deep

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learning-based approach, specifically a deep denoising autoencoder (DDAE), is employed to remove industrial noise. This noise reduction approach is adaptive across various levels of industrial noise. The denoised audio data are then used to train well-established, unsupervised, anomaly detection models, including an isolation forest (IF), a one-class support vector machine (OCSVM), and a local outlier factor (LOF). The performance of the proposed methodology was validated using the malfunctioning industrial machine investigation and inspection (MIMII) dataset [14]. The main contributions of the study may be summarized as follows:

- Audio data preprocessing technology is employed to enhance processing speed and performance.
- A noise reduction method appropriate for various industrial environments is introduced.
- The application and flexibility of the proposed methodology are evaluated in real-world scenarios, thus using audio data from different types of machinery, including fans, pumps, and slide rails, in the MIMII dataset. The results demonstrate high performance, a low inference time, and low memory usage.

II. RELATED WORK

A. DENOISING METHODS

In industrial settings, environmental sounds can readily contaminate audio data, adversely affecting machine failure detection. Several methods have been developed to reduce noise mixed with audio data. Common approaches use various types of filters, such as band-pass, high-pass, low-pass, and median filters [15], to select specific frequencies or amplitudes. These techniques are easy to implement and are widely used across various applications. However, they are primarily effective in environments with low levels of noise and are associated with a risk of inadvertent elimination of essential signals from audio data with high noise levels.

The use of deep learning for noise removal can overcome these problems [16]. Specifically, an autoencoder (AE) encodes input data to extract representative features and then uses these features to reconstruct the input data [17]. Building on this concept, a denoising AE has been employed to reconstruct noise-free signals from inputs to which noise was intentionally added [18]. When a series of encoder and decoder layers are employed, the configuration is termed a DDAE [19]. The DDAE has demonstrated superior noise removal performance across various domains, including speech recognition [19], [20] and reduction of random noise in seismological data [21], improving speech quality and noise reduction compared to traditional methods [20]. However, most deep-learning approaches are designed to handle the low-noise data of laboratory environments and often do not adequately consider varying noise levels [22]. Therefore, this study proposes a methodology utilizing the DDAE, which effectively reduces noise, and adapts it to serve across environments with various levels of industrial noise.

B. FEATURE EXTRACTION

Several methods have been used for feature extraction from recorded audio data. Peeters [23] outlined a comprehensive approach that extracts 166 features from such data. These features include the zero crossing rate, which quantifies the frequency at which a signal changes from positive to negative and vice versa; autocorrelation, reflecting the degree of similarity between a signal and a delayed version of that signal after various time lags; a mel-frequency cepstral coefficient (MFCC), which summarizes the spectral shape using a compact set of coefficients; and various measures of spectral distribution, including the mean, standard deviation (std), variance, skewness, and kurtosis. Rabaoui et al. [24] have contributed to the field by extracting a broad array of features, including time- and frequency-domain attributes, linear and cognitive linear predictions, cepstral features, and wavelet-based parameters from audio data.

C. ANOMALY DETECTION

Many authors have focused on unsupervised anomaly detection based on sound using various feature extraction methods. Nam et al. [25] transformed audio data acquired by a surface-mounted device-assembly machine into spectrograms and employed an AE for anomaly detection; this provided effective performance for both single and multiple products. Duman et al. [26] evaluated the performance of a convolutional autoencoder (CAE), an OCSVM, and a CAE-OCSVM in terms of converting audio data into mel-spectrograms, and confirmed that the CAE afforded superior performance in the plant industry. Becker et al. [27] employed long short-term memory (LSTM) to detect printer anomalies using features extracted based on MFCCs and the mel-filter bank energy. König et al. [28] detected anomalies in bearing systems by extracting features via continuous wavelet transformation using an AE. Pereira et al. [29] extracted MFCC features, gammatone frequency cepstral coefficients, and linear coding coefficients predictive of feature extraction, and employed an LSTM-AE for anomaly detection inside a vehicle.

The introduction of the MIMII dataset [14] has spurred the development of novel, unsupervised, anomaly detection methods tailored to the audio datasets of various factory machines. Van Truong et al. [30] extracted and combined the features of various types of audio data, including MFCC, chroma, mel-spectrogram, spectral contrast, and tonnetz features and then used a fully connected U-Net AE architecture for anomaly detection. Dohi et al. [31] converted audio data into mel-spectrograms and performed anomaly detection using self-supervised density estimation methods, namely the Glow add., Glow aff., MAF add., and MAF aff. Sha et al. [32] augmented audio data and extracted features using a discrete Fourier transform, and then detected anomalies using the GRLNet adversarial learning method. Di Fiore et al. [33] extracted features using a one-hot encoding method based on mel-spectrograms and machinery identifiers (IDs), and

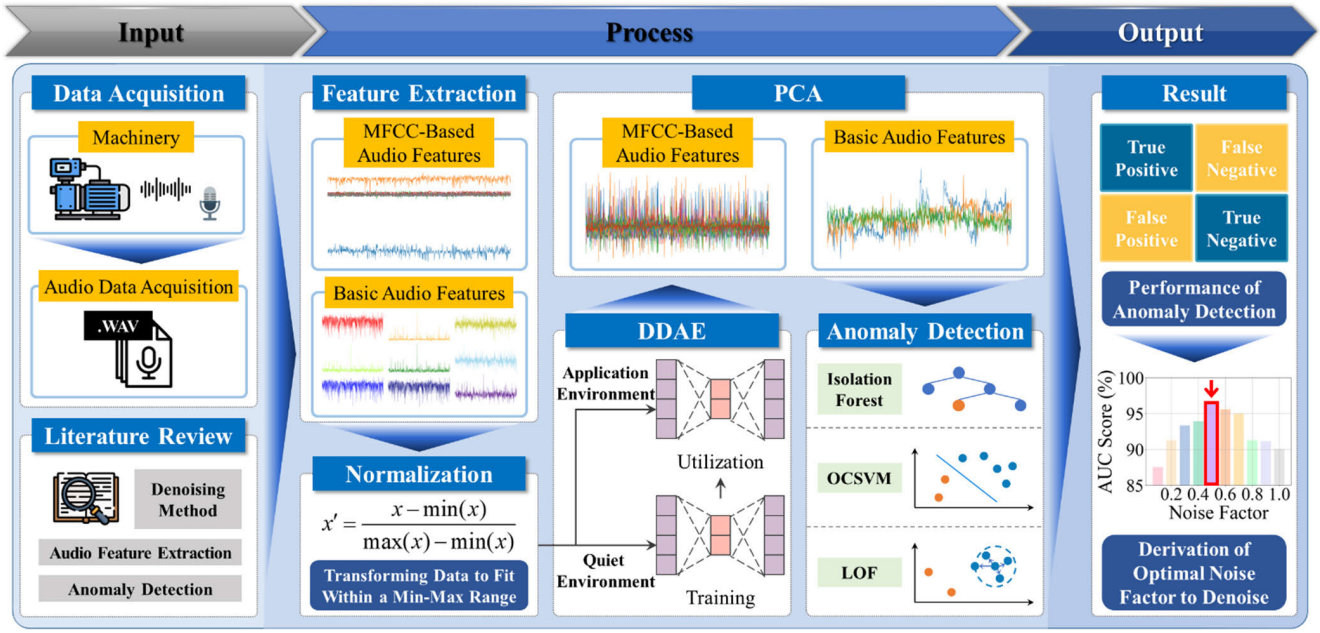

FIGURE 1. Overall framework of the proposed methodology for sound-based anomaly detection in noisy industrial environments. This framework consists of three main stages: input, process, and output.

detected machinery anomalies using two models, thus an ID-conditioned LSTM autoencoder (IDC-LSTM-AE) and an ID-conditioned convolutional autoencoder (IDCCAE).

Although many studies have focused on model performance, model training methods, and data preprocessing for unsupervised anomaly detection using sound, few studies have considered the practical applicabilities of such methods in the field. Therefore, this study presents an unsupervised methodology for anomaly detection that can be effectively applied in real-world settings. To verify the effectiveness of the proposed methodology, the system was compared with several high-performing anomaly detection models, including a Fully Connected U-Net, Glow aff., GRLNet, IDCCAE, and IDC-LSTM-AE (see Section IV).

III. THE SOUND-BASED ANOMALY DETECTION METHOD

The sound-based anomaly detection method of this study extracts MFCC-based and basic audio features from audio data. The MFCC-based audio features capture the overall shape of the audio signal, whereas the basic audio features capture the temporal information essential for the detection of machine malfunctions. In particular, Logan [34] observed that when MFCCs were used for preprocessing during feature extraction, this did not adversely affect the performance of audio identification algorithms. Both MFCC-based and basic audio features are computationally efficient, thus enabling real-time analysis and anomaly detection using sound. Such efficiency is particularly vital in industrial plants, wherein quick response times are paramount. Furthermore, the combination of MFCC-based and basic audio features facilitates comprehensive monitoring of a broad spectrum of machine conditions.

Given these advantages, this study extracts both MFCC-based and basic audio features. The extracted features are normalized and, using audio data recorded in quiet environments, are then employed to train a DDAE. Audio data recorded in industrial environments wherein anomaly detection is required are denoised using the trained DDAE. The denoised features are then subjected to dimensionality reduction via principal component analysis (PCA). The reduced data are next used to train well-established, unsupervised, anomaly detection models, including the IF, OCSVM, and LOF models, and the performances subsequently evaluated. This enabled discrimination of normal and abnormal machinery states using an optimal denoising method tailored to the noise levels of the specific industrial environment where anomaly detection is required. Fig. 1 illustrates the framework of the proposed methodology.

A. DATASET AND DATA PREPROCESSING

In this study, the MIMII dataset [14] was utilized; this contains audio data recorded during normal and abnormal operation of various factory machines in real industrial settings, including fans, pumps, and slide rails. Table 1 describes the operations of such machines and the anomalous conditions that may be encountered.

The dataset was collected using a circular microphone array consisting of eight distinct microphone elements, as illustrated in Fig. 2. The microphone array, with a diameter of 68 mm, was positioned 500 mm from the machine. The fan was closest to element 5, the pump to element 3, and the slide rail to element 7. In this study, single-channel audio data recorded by the element nearest to each type of machine are employed, rather than multi-channel audio data recorded

TABLE 1. Machine descriptions with some anomalous conditions.

Machine type	Operations	Anomalous conditions
Fan	Provision of continuous gas or air flows in factories.	Imbalance, voltage change, clogging, etc.
Pump	Suction from/discharge to a water pool.	Leakage, contamination, clogging, etc.
Slide rail	Repeated sliding at different speeds.	Rail damage, loose belt, no grease, etc.

by all microphones. These single-channel audio data were chosen because they afforded the clearest signals, thus those with the least amount of noise.

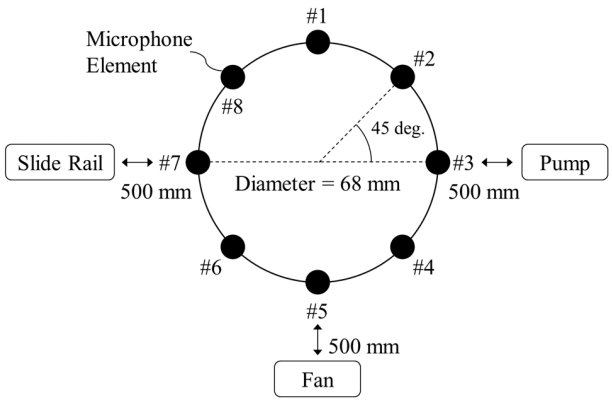

FIGURE 2. Circular microphone array. The microphone elements are arranged in a circular configuration with a diameter of 68 mm. Elements are numbered from 1 to 8, with adjacent elements separated by 45 degrees.

Each machine was recorded for 10 s at a sampling rate of 16 kHz, with recordings for each type of machine obtained in separate sessions. The four IDs are ID 00, ID 02, ID 04, and ID 06, each representing different products of each type of machinery. The dataset was designed to handle actual factory environments and thus included data with signal-to-noise ratios (SNRs) of 6 dB, 0 dB, and −6 dB, with equivalent amounts of data for each SNR level. The SNR formula is (1):

$$\mathrm{SNR}_{\mathrm{dB}}=10\log_{10}\left(\frac{P_{\mathrm{signal}}}{P_{\mathrm{noise}}}\right) \tag{1}$$

where P_{signal} and P_{noise} are the powers of the signal and noise, respectively. In this study, an SNR of 6 dB was assumed to reflect a quiet environment, thus that of an office or laboratory, or rooms where single machines operate in isolation. In such cases, the signal power is approximately four-fold that of noise; the target sound is easily distinguishable from background noise. An SNR of 0 dB was assumed to represent a low-noise environment, thus that of a factory production line or scenarios in which multiple machines operate concurrently. In such a case, the signal power is equal to that of noise, rendering it difficult to easily identify the target sound. An SNR of −6 dB was assumed to represent a high-noise environment, similar to a construction site. In such a case, the noise power is approximately four-fold that of the signal power, rendering it very difficult to separate and identify the target sound.

The proposed methodology includes training a DDAE with audio data recorded in the quiet environment. The trained DDAE is then used to denoise audio data recorded in a specific industrial environment wherein anomaly detection is required, thus environments with various noise levels, including quiet, low-noise, and high-noise environments. The audio data recorded in the quiet environment, and then used to train the DDAE, constitutes 20% of the total dataset, corresponding to approximately 20 to 50 min of recorded audio. To train and evaluate the anomaly detection model, the remaining data were split into 70% for training and 30% for testing. When performing unsupervised anomaly detection, it is essential to train the model using audio data from normal mechanical devices; therefore, the training data consisted entirely of such data. Table 2 lists the number of samples used to train the DDAE, and to train and test the anomaly detection models; the sample numbers were the same for each SNR.

TABLE 2. Sample counts used for DDAE training and anomaly detection training and testing.

Machine	ID	DDAE (SNR 6 dB)	Anomaly detection (SNRs of 6 dB, 0 dB, −6 dB) Training	Testing
Fan	00	285	794	339
	02	276	769	330
	04	278	773	330
	06	277	770	329
Pump	00	231	643	275
	02	225	624	267
	04	162	448	192
	06	229	637	272
Slide rail	00	287	797	340
	02	268	747	320
	04	144	398	170
	06	126	348	149

1) MFCC-BASED AUDIO FEATURES

The MFCC is widely used for feature extraction during audio signal processing. The use of MFCCs requires segmentation of audio data into small frames, followed by use of the fast Fourier transform (FFT) to convert the time domain into frequency components, thereby generating a spectrum. Subsequently, the mel-filter bank is employed to analyze the frequency components of the audio signal; the resolution is finer in the low-frequency than the high-frequency region. Finally, the MFCC features are extracted using the discrete cosine transform (DCT). Fig. 3 illustrates the overall MFCC extraction process.

The MFCC thus allows the extraction of frequency-time domain features, enabling the identification of specific frequencies in audio data at particular time intervals. In this study, 128-dimensional MFCC vectors were extracted, and the mean value of the corresponding MFCCs for each dimension was used to extract MFCC-based audio features. Such

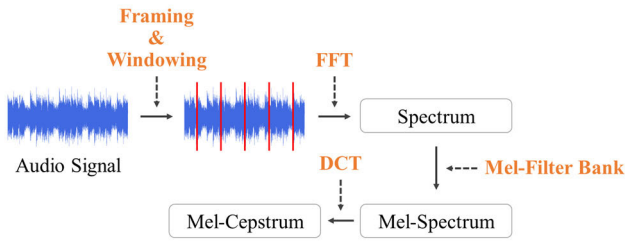

FIGURE 3. Process of extracting MFCCs. The audio signal is first subjected to framing and windowing, followed by the FFT to produce the spectrum. The spectrum is then converted to the mel-spectrum using a mel-filter bank, and finally, the mel-cepstrum is obtained by applying the DCT.

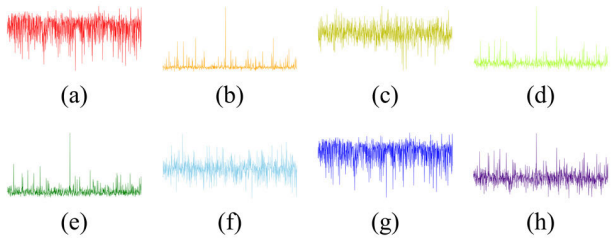

FIGURE 4. Graphical representations of basic audio features illustrating key statistical and structural properties of audio signals: (a) RMS, (b) kurtosis, (c) skewness, (d) peak-to-peak, (e) crest factor, (f) mean, (g) std, and (h) entropy.

features are highly robust against noise and help reduce data dimensionality.

2) BASIC AUDIO FEATURES

Basic audio feature extraction entails the computation of straightforward yet insightful statistical measures derived from the audio signals. This yields critical information, including sound pressures; and kurtosis, skewness, and asymmetry data. The root mean square (RMS) captures the mean signal magnitude over a defined interval, effectively serving as a surrogate of the average sound pressure level. Kurtosis assesses the prominence of tails in the signal probability distribution, whereas skewness is a metric of distributional asymmetry. The peak-to-peak metric is the difference between the maximum and minimum signal amplitudes, and thus captures the entire amplitude range. The crest factor, thus the ratio of the peak signal amplitude to the RMS, indicates pulsatility. Entropy measures signal unpredictability, thus the extents of inherent irregularity and chaos.

Given the multifaceted characteristics of audio signals, it is imperative to extract the following basic audio features: RMS, kurtosis, skewness, peak-to-peak, crest factor, mean, std, and entropy. The selected features effectively encapsulate the core characteristics of the audio signals of our study. Other potential features were excluded because they either imposed a high computational burden or were redundant. Collectively, the chosen features afford a profound understanding of the attributes and patterns embedded in the audio signal. Fig. 4 presents graphical visualizations of these features extracted from audio data in the MIMII dataset.

3) NORMALIZATION

MFCC-based and basic audio features contain different types of information that may use different units or exhibit varying ranges of values. In such cases, min-max normalization is performed to ensure that all features are on the same scale. The normalization equation is (2):

$$x'_f = \frac{x_f - \min(x_f)}{\max(x_f) - \min(x_f)} \tag{2}$$

where x_f is the feature value and x'_f the normalized feature value. Both MFCC-based and basic audio features, extracted from audio data recorded in a quiet environment, are normalized. The DDAE is then trained on these features for denoising purposes. Subsequently, both MFCC-based and basic audio features, extracted from audio data recorded in a specific industrial environment where anomaly detection is required, are normalized and then denoised using the trained DDAE.

B. THE DEEP DENOISING AUTOENCODER

Vincent et al. [35] developed a denoising AE, an enhancement of a traditional AE, by deliberately adding noise to the input data. The denoising AE trains using these corrupted samples to reconstruct the clean inputs, marking an advance over the traditional AE. The structure of the denoising AE is depicted in Fig. 5.

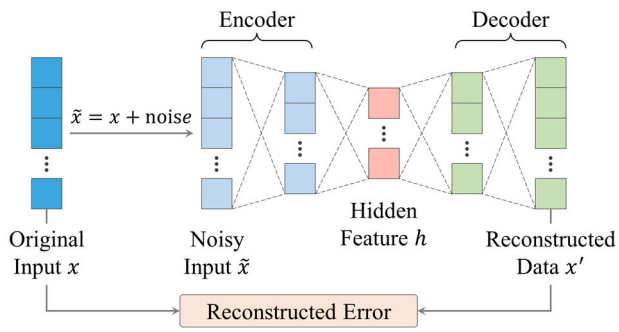

FIGURE 5. Structure of a denoising AE. The original input data is corrupted with noise to produce the noisy input data. The encoder maps the noisy input data to a hidden feature, and the decoder reconstructs the input from the hidden feature to generate the output data. The reconstruction error is the difference between the original input data and the reconstructed data.

To create a noisy input $\tilde{x}$, noise is added to the original input x. In this study, Gaussian white noise was introduced, as it well approximates the noise of various real-world environments. The added noise is described by (3):

$$\begin{cases} x \sim N(\text{mean}, \sigma^2) \\ \text{noise} \sim N(0, (\text{NF} \times \sigma)^2) \end{cases} \tag{3}$$

where σ is the std of x and NF the noise factor of this study. On evaluation of method performance at different noise

factors (0.1, 0.2, …, 1), the noise factor appropriate for the noise that occurs in real industrial environments is identified.

The generated noisy input $\tilde{x}$ is fed into the AE, wherein encoding-decoding takes place. During this process, the noisy input $\tilde{x}$ is mapped to a hidden feature h via the encoder, as described by (4), and the h is then reconstructed the reconstructed data x' via the decoder, as described by (5).

$$h = f_\theta(\tilde{x}) = a(W\tilde{x} + b) \tag{4}$$

$$x' = g_{\theta'}(h) = a\left(W'h + b'\right) \tag{5}$$

In (4) and (5), a is the activation function (ReLU was used for all layers except the output layer, which employed the sigmoid function); W and W' were the weights and b and b' the biases, with $\theta = \{W, b\}$ and $\theta' = \{W', b'\}$.

A loss function typically employs the squared error as shown in (6) to determine reconstructed data loss during AE operation [36], [37]. Subsequently, the model is optimized to minimize this loss, as described by (7):

$$L(x, x') = \|x - x'\|^2 \tag{6}$$

$$\theta^*, \theta'^* = \arg\min \frac{1}{n} \sum_{i=1}^{n} L\left(x^{(i)}, x'^{(i)}\right) \tag{7}$$

In (7), $x^{(i)}$ and $x'^{(i)}$ are the i-th elements of x and x', respectively. When the denoising AE features multiple hidden layers, it is termed a DDAE [19]. The additional layers allow the DDAE to discern and eliminate more subtle noise from the input data [38]. In this study, the DDAE served as a noise removal filter that denoised features extracted from audio data with industrial noise.

MFCC-based audio features use an undercomplete DDAE, and basic audio features employ an overcomplete DDAE. The dimensions of hidden layers in an under-/over-complete DDAE are smaller/larger, respectively, than that of the input layer. As MFCC-based audio features are of 128 dimensions, the use of an undercomplete DDAE more efficiently represents data while preserving essential information. MFCC-based audio features compress the audio frequency contents; an undercomplete DDAE usefully extracts information from such compressed features. However, as the basic audio features are of only 8 dimensions, the use of an undercomplete DDAE would oversimplify the network and thus fail to adequately represent the data complexity. Therefore, an overcomplete DDAE is used to map the input data to a higher-dimensional subspace, thereby enhancing representation via additional lateral connections [39].

To determine the optimal DDAE topology, the architecture was systematically manipulated in several stages. We began by adjusting the number of neurons per layer; this critically affects the capacity of a model to effectively encode and decode input data. Neuron counts of 8, 16, 32, 64, 128, 256, and 512 were tested to ascertain their impacts on model performance. Subsequently, various topologies were examined to optimize the model parameters. Following these initial tests, further modifications involved concurrent adjustments of both the number of layers and the number of neurons per layer using testing configurations of 1 to 15 layers to identify the optimal layer depth for effective noise reduction. Alongside such architectural adjustments, hyperparameters such as the learning rate, batch size, and the number of training epochs were iteratively optimized based on their impacts on the convergence speed and loss minimization. This iterative process continued until the best network topology and optimal DDAE hyperparameters were attained. In this study, the mean squared error serves as the loss function when assessing the outcomes. Based on these experiments, the optimal network topology is illustrated in Fig. 6, and the best DDAE hyperparameters are presented in Table 3.

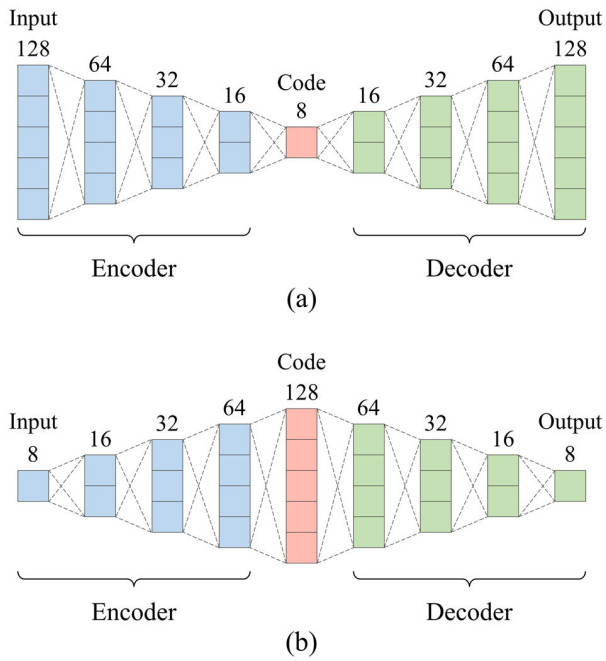

FIGURE 6. Proposed DDAE network architectures by type of features: (a) an undercomplete DDAE for MFCC-based audio features; (b) an overcomplete DDAE for basic audio features.

TABLE 3. Hyperparameters used to train the DDAE.

Hyperparameter	Value	Remark
Learning rate	0.001	Determines by how much the weights of the model are adjusted.
Batch size	4	The number of training sequences used in each iteration.
Optimizer	Adam	An algorithm employed to update the weights of the model.
Epochs	500	An epoch is one complete presentation of all training data to the model.

C. PRINCIPAL COMPONENT ANALYSIS

After denoising audio data recorded in an industrial environment where anomaly detection is required, PCA is performed. PCA, first described by Pearson [40], is a multivariate data

analysis technique that reduces data dimensionality, thereby simplifying and compressing the dataset. The number of principal components is determined using the minimum average partial (MAP) test proposed by Velicer [41], which calculates the average of the squared partial correlation coefficients among k variables, given the first m principal components, using a partial correlation matrix. The m corresponding to the minimum value is the number of principal components. The MAP test equation is (8):

$$\mathrm{MAP}_m = \sum_{i=1}^{p} \sum_{\substack{j=1 \\ i \neq j}}^{p} \frac{{r_{ijm}^{*}}^{2}}{p(p-1)} \tag{8}$$

where ${r_{ijm}^{*}}^{2}$ is the value obtained by subtracting the proportion of the interaction between the i-th principal component and all other principal components from the covariance ratio of the proportion of the variance explained by the i-th and j-th principal components; and k is the number of variables.

The MFCC-based and basic audio features obtained when PCA was used to obtain the optimal number of principal components, as determined by the MAP test, are shown in Fig. 7. After PCA, these features were combined to create the training data. This approach effectively reduces data dimensionality while preserving essential information, thus enhancing the efficiency of both learning and analysis.

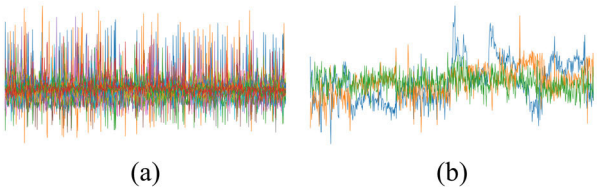

FIGURE 7. Graphs of features after PCA transformation: (a) MFCC-based audio features; (b) basic audio features.

D. ANOMALY DETECTION

The MFCC-based and basic audio features extracted from audio data recorded in a specific industrial environment where anomaly detection is required undergo normalization, denoising via the DDAE, and dimensional reduction via PCA. These processed data yield all anomalies when input to three well-established, unsupervised, learning-based anomaly detection models. Each model employs a distinct approach toward outlier detection; the models are IF, OCSVM, and LOF.

The IF operates using a tree structure to isolate data points, predicated on the assumption that outliers are more readily separable in regions of sparse data. IF constructs a binary tree by recursively splitting the data using randomly generated one-dimensional lines at each node, effectively isolating points via tree pruning. The tree is built by randomly selecting attributes, but assembly of a forest of multiple trees mitigates the randomness of the pruning criteria. At each node, when a data point is chosen, the algorithm measures the distance from that point to the splitting criterion. A shorter distance implies that the point is likely not an outlier; a longer distance indicates the opposite. Thus, outliers are detected. In this study, all parameters of the IF model were those initially proposed by Liu et al. [42].

The OCSVM is an anomaly detection model that applies the support vector machine framework to a single-class setting; the focus is on identification of the region of data normality. Outliers are data points that fall outside this region. By projecting input data into a higher-dimensional space, OCSVM detects outliers by establishing an optimal boundary that separates the normal data distribution from the outliers, thus maximizing the distance from the origin to the nearest data point in the transformed feature space. In this study, all parameters of the OCSVM model were those initially proposed by Schölkopf et al. [43].

The LOF is a density-based anomaly detection method that assesses the local density deviation of a given data point relative to that of its neighbors. Data points exhibiting significantly lower local densities than their neighbors are flagged as outliers. This enables LOF to detect local anomalies in datasets characterized by variable densities across different regions, thus the ability to detect outliers that might not be discernible on global analysis. In this study, all parameters of the LOF model were those initially proposed by Breunig et al. [44].

Using these three models, it is possible to discriminate between normal and abnormal machinery states and derive the optimal noise factor for the various levels of industrial noise. The performance of our proposed methodology was experimentally evaluated using these models; we present an optimal denoising method tailored to the noise levels of the industrial environments wherein anomaly detection is required.

IV. EXPERIMENTAL EVALUATION

In this section, the derivations of optimal noise factors for various levels of industrial noise, and the anomaly detection performance and execution time, are presented. The performance evaluation metric employed is the area under the curve (AUC) of the receiver operating characteristic (ROC). The ROC graph features the false-positive rate on the x-axis and the true-positive rate on the y-axis; the AUC is the area under the ROC curve. An AUC close to 1 indicates that a model perfectly discriminates between normal and anomalous conditions, and is particularly useful when fairly assessing the performances of models that handle imbalanced datasets.

A. COMPARISON OF DIFFERENT ANOMALY DETECTION MODELS BY THE SNR

The performance of the proposed methodology across various machines, including fans, pumps, and slide rails, at SNRs of 6 dB, 0 dB, and −6 dB, using different noise factors, is presented in Table 4. The conventional methodology of Table 4 refers to anomaly detection without denoising. This comprehensive overview facilitates a direct comparison of

TABLE 4. AUC scores (%) of the proposed methodology and of conventional methods without denoising (Bold indicates the best valueS).

SNR	Machine	Model	Conventional	Proposed (noise factor)									
				0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1
6 dB	Fan	IF	96.17	94.57	**96.97**	94.4	94.04	93.75	93.14	92.8	91.39	91.28	92.81
		OCSVM	76.35	**76.37**	75.61	74.53	74.09	73.52	72.39	71.93	73.03	72.72	72.7
		LOF	95.35	94.82	**98.37**	96.53	96.45	93.1	92.04	92.62	91.52	93.44	91.4
	Pump	IF	**95.29**	93.33	94.16	94.03	93.87	93.54	93.29	92.51	89.52	87.54	85.6
		OCSVM	73.08	76.85	76.39	74.97	75.88	76.55	**76.87**	76.53	76.36	75.33	74.4
		LOF	91.48	94.25	**96.33**	94.95	95.49	93.52	93.28	92.33	89.73	85.76	82.69
	Slide rail	IF	93.02	92.94	**94.24**	92.11	92.71	92.34	92.32	91.67	89.95	90.24	86.04
		OCSVM	71.97	78.23	**80.16**	79.49	76.53	76.54	76.85	76.11	76.63	76.8	72.99
		LOF	88.16	93.78	**94.22**	91.9	92.89	92.64	93.71	93.6	93.4	91.51	86.76
0 dB	Fan	IF	87.83	87.52	91.25	93.33	93.92	**96.51**	95.57	94.97	91.25	91.12	90.06
		OCSVM	72.96	74.75	75.59	75.79	79	**79.55**	78.87	75.38	72.89	72.63	71.13
		LOF	83.56	86.16	89.09	90.75	94.16	**96.48**	95.31	95.59	92.59	93.74	92.42
	Pump	IF	92.14	92.19	92.97	93.85	93.5	**95.31**	92.71	91.78	87.4	86.62	85.83
		OCSVM	73.19	74.85	75.43	75	75.94	**76.63**	76.39	75.66	73.64	74.16	73.38
		LOF	81.63	91.04	92.9	93.53	93.91	**94.6**	93.94	93.09	87.17	87.66	81.16
	Slide rail	IF	88.85	91.02	93.45	93.7	93.58	**94.88**	90.83	90.21	88.86	88.38	86.58
		OCSVM	74.11	75.26	75.59	**76.35**	76.07	75.83	73.73	72.47	74.95	72.56	70.03
		LOF	75.22	91.17	91.12	91.48	91.64	**92.9**	91.27	90.34	91.78	87.75	84.59
-6 dB	Fan	IF	70.66	83.02	84.12	88.08	89.59	87.11	89.26	**90.11**	87.21	86.55	86.1
		OCSVM	65.77	73.88	71.76	73.69	73.95	73.71	73.45	**76.22**	75.6	72.36	72.07
		LOF	67.7	76.61	81.99	87.44	85.62	88.15	89.64	**90.4**	84.1	85.7	86.97
	Pump	IF	72.73	83.37	85.37	86.55	87.64	86.5	86.91	**88.8**	84.53	81.12	78.52
		OCSVM	71.18	74.14	73.45	73.81	73.94	73.88	74	**74.44**	71.97	72.49	71.36
		LOF	67.44	77.46	86.82	88.95	91.13	**91.35**	90.04	90.5	84.67	83.71	78.88
	Slide rail	IF	79.17	85.57	89.17	89.97	89.6	89.67	92.62	**93.38**	91.96	87.57	78.44
		OCSVM	69.07	74.92	75.29	75.76	75.26	75.08	75.44	**78.3**	73.88	72.57	71.02
		LOF	65.97	82.67	85.67	87.93	89.38	88.07	88.51	89.97	**91.92**	87.83	81.48

the anomaly detection capability of each model, with the highest AUC highlighted in bold. In most cases, the IF afforded the highest average AUC. The optimal noise factor, thus that yielding the highest average AUC, was 0.2 in a quiet environment (SNR 6 dB), 0.5 in a low-noise environment (SNR 0 dB), and 0.7 in a high-noise environment (SNR −6 dB).

B. COMPARISON BETWEEN THE PROPOSED METHOD AND PREVIOUS STUDIES

The results obtained when comparing the performance of the proposed methodology with the four previous studies mentioned in Section II are presented in Table 5. To ensure a fair comparison with the other studies, only the results obtained at an SNR of 0 dB were considered. At this SNR, noise was removed using a DDAE with a noise factor of 0.5, and anomaly detection was performed using the IF, which demonstrated superior AUC performance for all of the fan, pump, and slide rail. Specifically, the average AUC increased by 27.89% for the fan, 16.68% for the pump, and 8.36% for the slide rail.

The proposed methodology offers rapid training and inference speeds, high performance, and low memory usage. The average time and memory usage required for anomaly detection are presented in Table 6. Using the proposed methodology and 1 MiB of memory, training requires an average of approximately 2 min and 23 s, and the inference is complete in about 0.13 s. Compared to previous studies that considered real-life applicability, both memory usage, and the training and inference times, were notably enhanced.

TABLE 5. Performance comparison between the proposed methodology and previous studies of anomaly detection in terms of the AUC metric (%) (bold indicates the best values).

Author	Model	Year	Machine		
			Fan	Pump	Slide rail
Proposed		2024	**96.51**	**95.31**	**94.88**
Truong et al. [30]	Fully Connected U-Net	2021	80.06	85.97	83.10
Dohi et al. [31]	Glow aff.	2021	74.80	83.10	94.80
Sha et al. [32]	GRLNet	2022	86.62	85.31	80.85
Di Fiore et al. [33]	IDCCAE	2022	71.05	76.63	90.99
	IDC-LSTM-AE	2022	67.66	78.29	82.62

TABLE 6. Evaluation of the proposed methodology in terms of training time, inference time, and memory usage (bold indicates the best values).

Model	Machine	Training	Inference	Memory (MiB)
Proposed	Fan	**2 min 52 s**	**0.16 s**	**1**
	Pump	**2 min 10 s**	**0.11 s**	
	Slide rail	**2 min 6 s**	**0.11 s**	
IDCCAE [33]	Fan	1 h 58 min 29 s	3.04 s	8
	Pump	1 h 26 min 51 s	3.19 s	
	Slide rail	2 h 2 min 40 s	2.78 s	
IDC-LSTM-AE [33]	Fan	6 min 44 s	14.70 s	64
	Pump	3 min 49 s	15.40 s	
	Slide rail	5 min 9 s	14.6 s	

C. DISCUSSION

The proposed methodology demonstrates exceptional resistance to overfitting in environments with very variable noise levels. This is evidenced by the consistent performance across

different SNRs of 6 dB, 0 dB, and −6 dB. For example, at an SNR of −6 dB, where noise levels are significantly high, setting the noise factor to 0.7 yielded the highest average AUC. Such adjustment enables the DDAE to effectively filter out noise-specific features, preventing the model from learning noise as a signal, which is a common pitfall that causes overfitting.

The generalization of the proposed methodology across various machine types and industrial environments was demonstrated via application to diverse mechanical components, including fans, pumps, and slide rails. These components often operate under different conditions, and are exposed to various types of operational and environmental noise. Despite these variations, compared to previous studies, the proposed methodology achieved notable improvements in anomaly detection, thus a 27.89% increase in the AUC for fans, 16.68% for pumps, and 8.36% for slide rails at an SNR of 0 dB.

Additionally, the proposed methodology offers rapid training and inference speeds, superior performance, and low memory usage. These attributes not only underscore the potential for on-site applicability but also the capacity to conduct real-time anomaly detection in factory equipment using edge computing. The proposed methodology was implemented using the Python programming language, with the Scikit-learn and PyTorch libraries on a single GPU, an NVIDIA GeForce RTX 3080 Ti, and a Windows workstation environment that leveraged CUDA 12.3.

V. CONCLUSION

Our goal is to detect anomalous machinery conditions rapidly and accurately using audio data that include various levels of industrial noise. This would greatly mitigate potential problems in actual industrial settings, including reduced productivity, safety hazards, and increased costs. The literature review explored the various features that can be extracted from sound and case studies of unsupervised anomaly detection using sound. Subsequently, a methodology for sound-based anomaly detection, designed to ensure quick and accurate identification of abnormal machine states within audio data containing various levels of industrial noise, was proposed. The methodology evidenced a rapid processing speed and high accuracy, rendering it practically applicable in industrial environments.

The proposed methodology extracts MFCC-based and basic audio features from recorded audio data. Features extracted from audio data recorded in a quiet environment are normalized and then used to train the DDAE. Features extracted from audio data recorded in an industrial environment, where anomaly detection is required, are denoised using the trained DDAE. These denoised features are then subjected to dimensional reduction via PCA, after which they are analyzed using well-established, unsupervised, learning-based, anomaly detection models. The methodology was validated using audio data from real industrial machines, which demonstrated that the IF model afforded the highest average AUC. The optimal noise factors were 0.2 in quiet environments (SNR 6 dB), 0.5 in low-noise environments (SNR 0 dB), and 0.7 in high-noise environments (SNR −6 dB); denoising was then effective. The main advantages of the proposed methodology may be summarized as follows:

- High and consistent performance across different SNRs of 6 dB, 0 dB, and −6 dB.
- Generalization across various machine types and industrial settings.
- Rapid inference time, averaging 0.13 s, with a low memory usage of 1 MiB.
- An average improvement of 18% in AUC performance compared to previous methods.
- Cost-effectiveness and suitability for edge computing and on-site applications.

Our main contributions are the data preprocessing techniques employed to enhance processing speed and performance, and the tailored DDAE techniques that denoise audio data recorded in specific industrial environments where anomaly detection is required. Our approach overcomes the training time and accuracy problems associated with conventional anomaly detection methods. Moreover, the proposed methodology can be implemented at low cost, thereby increasing the potential for edge computing and on-site applicability. In real-world industrial settings, this technology not only will guarantee cost and time efficiencies but also is applicable to diverse industrial scenarios.

To further enhance the robustness of anomaly detection, future experiments involving the integration of advanced machine learning techniques are planned. The aim is to broaden model applicability across ever more diverse industrial environments.

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